Question 1

Explain the difference between EPC (4G) and 5GC (5G).

EPC is the older 4G core that uses hardware based nodes, while 5GC is fully software based and built on a service based architecture that's way more flexible and scalable.

Question 2

Why is network slicing important in 5G? Provide real-world examples.

Network slicing lets carriers create virtual networks tailored to specific needs, like one slice for autonomous cars that need ultra low latency and another for IoT sensors that just need low bandwidth.

Question 3

What is Multi-access Edge Computing (MEC), and how does it reduce latency?

MEC basically pushes computing power closer to the user instead of routing everything to a distant data center, so things like AR apps or real time gaming don't lag as much

Question 4

Describe the role of AI in dynamic resource allocation.

AI monitors network traffic in real time and automatically shifts resources like bandwidth or processing power to where they're needed most, so nothing gets overloaded or wasted.

Question 5

In SBA, what is the role of the Network Repository Function (NRF)?

The NRF acts like a directory for the 5G core when one network function needs to talk to another, it checks in with the NRF to find and connect to the right service.